DATA COMMUNICATION FUNDAMENTALS

Complete 50-Page Course Textbook

Beginner → Intermediate

Theory • Networking Concepts • Transmission • Protocols • Exercises • Assessments

A structured learning resource covering how digital information is represented, transmitted, received, and controlled across communication systems.

Course Overview

Purpose: This course introduces the principles of data communication and computer networking. It covers signals, transmission media, encoding, multiplexing, switching, error control, network architectures, protocols, and practical network design.

Learning outcomes: Learners should be able to explain data communication concepts, identify communication components, compare transmission methods, describe network topologies and media, explain OSI/TCP-IP layers, understand error detection and correction, and design a basic network.

Table of Contents

- 1. Introduction to Data Communication
- 2. Components of Data Communication
- 3. Data and Information
- 4. Communication Signals
- 5. Analog and Digital Signals
- 6. Signal Characteristics
- 7. Bandwidth and Data Rate
- 8. Transmission Impairments
- 9. Guided Transmission Media
- 10. Unguided Transmission Media
- 11. Serial and Parallel Transmission
- 12. Synchronous and Asynchronous Transmission
- 13. Simplex, Half-Duplex and Full-Duplex
- 14. Baseband and Broadband Transmission
- 15. Digital-to-Digital Encoding
- 16. Analog-to-Digital Conversion
- 17. Digital-to-Analog Conversion
- 18. Analog-to-Analog Conversion
- 19. Modulation and Demodulation
- 20. Multiplexing
- 21. Switching Techniques
- 22. Circuit Switching
- 23. Packet Switching
- 24. Message Switching
- 25. Error Detection
- 26. Error Correction
- 27. Flow Control

- 28. Data Link Protocols
- 29. Network Topologies
- 30. LAN, MAN, WAN and PAN
- 31. OSI Reference Model
- 32. TCP/IP Model
- 33. Physical and Data Link Layers
- 34. Network and Transport Layers
- 35. Application Layer Protocols
- 36. IP Addressing Fundamentals
- 37. Ethernet and MAC Addresses
- 38. Wireless Communication
- 39. Network Devices
- 40. Communication Security
- 41. Quality of Service
- 42. Network Performance
- 43. Client-Server and Peer-to-Peer
- 44. Internet and Modern Networks
- 45. Network Troubleshooting
- 46. Network Design Case Study
- 47. Revision Guide
- 48. 60-Question Assessment and Answers
- 49. Practical Exercises
- 50. Glossary

Lesson 1: Introduction to Data Communication

Lesson Content

Data communication is the exchange of information between devices through a medium using agreed protocols. A basic model contains sender, receiver, message, medium, and protocol. Effective communication considers delivery, accuracy, timeliness, and acceptable variation in timing.

Exercise

Identify the five components in a video call.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 2: Components of Data Communication

Lesson Content

The sender originates data, the receiver accepts it, the message contains information, the medium carries the signal, and the protocol defines communication rules. Protocols can specify format, timing, addressing, sequencing, and error handling.

Exercise

Describe these components using an email example.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 3: Data and Information

Lesson Content

Data consists of symbols or values representing facts, measurements, text, images, sound, or other content. Information is interpreted data. Digital systems commonly represent data using binary digits. Common units include bits and bytes.

Exercise

Explain why text, photographs, and audio can all be transmitted digitally.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 4: Communication Signals

Lesson Content

A signal is a physical representation used to carry information. Copper commonly uses electrical signaling, fiber uses light, and wireless systems use electromagnetic waves. Signals must be suitable for their transmission medium.

Exercise

Give two examples of communication signals.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 5: Analog and Digital Signals

Lesson Content

Analog signals vary continuously. Digital signals use discrete levels or symbols. Digital communication supports processing, regeneration, coding, and error-control techniques, although real systems still interact with analog physical channels.

Exercise

List three differences between analog and digital signals.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 6: Signal Characteristics

Lesson Content

Amplitude describes signal magnitude; frequency describes cycles per second; phase describes relative position within a cycle; wavelength is the distance corresponding to one cycle. These characteristics influence communication design.

Exercise

Define amplitude, frequency, phase, and wavelength.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 7: Bandwidth and Data Rate

Lesson Content

Bandwidth can refer to a channel's supported frequency range. Bit rate is bits per second, while baud rate is symbols per second. A symbol may represent multiple bits. Achievable throughput depends on channel quality and system design.

Exercise

Explain bit rate versus baud rate.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 8: Transmission Impairments

Lesson Content

Attenuation reduces signal strength. Distortion changes signal shape. Noise adds unwanted energy. Appropriate media, shielding, amplification or regeneration, filtering, coding, and careful system design can reduce their effects.

Exercise

Name three impairments and one mitigation for each.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 9: Guided Transmission Media

Lesson Content

Twisted pair uses insulated copper conductors and is common in Ethernet. Coaxial cable uses a central conductor and shielding. Optical fiber carries light and provides high bandwidth with strong resistance to electromagnetic interference.

Exercise

Compare twisted pair, coaxial cable, and fiber.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 10: Unguided Transmission Media

Lesson Content

Wireless systems use electromagnetic waves. Radio supports many applications; microwave can support high-capacity links; infrared is generally short range and can be affected by obstacles. Wireless performance depends on propagation conditions and interference.

Exercise

List two advantages and two limitations of wireless communication.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 11: Serial and Parallel Transmission

Lesson Content

Serial transmission sends bits sequentially, while parallel transmission sends multiple bits simultaneously over multiple paths. Serial signaling is widely used for modern high-speed links because it can simplify wiring and support reliable long-distance transmission.

Exercise

Give five differences between serial and parallel transmission.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 12: Synchronous and Asynchronous Transmission

Lesson Content

Asynchronous transmission associates timing information with individual characters or small units, often using start and stop mechanisms. Synchronous transmission uses coordinated timing and structured blocks or frames and is efficient for continuous data.

Exercise

Compare synchronous and asynchronous transmission.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 13: Simplex, Half-Duplex and Full-Duplex

Lesson Content

Simplex permits one-way communication. Half-duplex permits two-way communication but not simultaneously. Full-duplex permits simultaneous two-way communication. Different applications choose modes based on requirements.

Exercise

Give one example of each mode.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 14: Baseband and Broadband Transmission

Lesson Content

Baseband systems transmit digital signaling directly through a channel or communication system. Broadband systems commonly support multiple frequency channels or use modulation over a wider frequency range. Ethernet LAN signaling is a common baseband example.

Exercise

Explain the basic difference.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 15: Digital-to-Digital Encoding

Lesson Content

Line coding maps digital data to digital signal patterns. Examples include NRZ and Manchester-style schemes. Encoding affects synchronization, bandwidth, signal transitions, and DC characteristics.

Exercise

Explain why synchronization matters.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 16: Analog-to-Digital Conversion

Lesson Content

Sampling measures an analog signal at regular intervals; quantization maps samples to discrete levels; encoding represents those levels digitally. For a band-limited signal, the sampling rate must be sufficiently high to avoid aliasing.

Exercise

Explain sampling, quantization, and encoding.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 17: Digital-to-Analog Conversion

Lesson Content

Digital information can control a carrier's amplitude, frequency, or phase. Common conceptual methods include amplitude shift keying, frequency shift keying, and phase shift keying. Modems and radio systems use related techniques.

Exercise

Name three basic digital-to-analog approaches.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 18: Analog-to-Analog Conversion

Lesson Content

Analog information can modulate a carrier through amplitude modulation, frequency modulation, or phase modulation. These methods change different properties of a carrier while preserving information in the resulting waveform.

Exercise

Differentiate AM, FM, and PM.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 19: Modulation and Demodulation

Lesson Content

Modulation places information onto a carrier suitable for transmission. Demodulation extracts the information from a received carrier. A modem historically combines these functions. Modulation supports frequency allocation, antenna design, and efficient channel use.

Exercise

Explain the roles of a modulator and demodulator.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 20: Multiplexing

Lesson Content

Multiplexing allows multiple signals to share a communication resource. FDM separates signals by frequency, TDM separates them by time, and WDM separates optical channels by wavelength.

Exercise

Compare FDM, TDM, and WDM.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 21: Switching Techniques

Lesson Content

Switching directs communication through interconnected networks. Circuit, packet, and message switching represent major conceptual approaches. They differ in resource reservation, delay, efficiency, and handling of bursty traffic.

Exercise

Why is switching necessary in large networks?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 22: Circuit Switching

Lesson Content

Circuit switching establishes a dedicated or reserved communication path for a session. Traditional systems involve setup, data transfer, and teardown. It can offer predictable resources but may waste capacity during idle periods.

Exercise

Compare circuit and packet switching.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 23: Packet Switching

Lesson Content

Packet switching divides data into packets that share network resources. Datagrams can be forwarded independently, while virtual-circuit approaches maintain logical forwarding state. Packet switching is efficient for bursty traffic.

Exercise

Explain how packet switching improves resource utilization.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 24: Message Switching

Lesson Content

Message switching uses store-and-forward transfer of complete messages. Intermediate nodes receive and store a message before forwarding it. It avoids a dedicated circuit but can introduce substantial delay and storage requirements.

Exercise

State two advantages and two limitations.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 25: Error Detection

Lesson Content

Transmission errors can alter bits. Parity provides a simple check; checksums use arithmetic summaries; CRC uses polynomial-based computation and is effective at detecting many common transmission errors.

Exercise

Why is error detection important?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 26: Error Correction

Lesson Content

Error detection identifies corruption; correction recovers intended data through redundancy or retransmission. Forward error correction adds redundancy, while ARQ uses acknowledgements and retransmissions. More redundancy consumes additional resources.

Exercise

Compare FEC and retransmission.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 27: Flow Control

Lesson Content

Flow control prevents a fast sender from overwhelming a slower receiver. Stop-and-wait sends a unit and waits for confirmation. Sliding-window methods allow multiple outstanding units within a defined window.

Exercise

Explain sliding-window flow control.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 28: Data Link Protocols

Lesson Content

The data link layer organizes bits into frames and supports local-link delivery. It can provide link-layer addressing, error detection, flow control, and retransmission depending on the protocol.

Exercise

List four data-link functions.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 29: Network Topologies

Lesson Content

Bus uses a shared path; star connects devices to a central point; ring forms a logical or physical ring; mesh provides multiple paths. Hybrid designs combine approaches. Each has cost, reliability, and management trade-offs.

Exercise

Compare four topologies.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 30: LAN, MAN, WAN and PAN

Lesson Content

PAN covers a very small personal area. LAN covers a limited location. MAN can span a metropolitan area. WAN connects networks over large geographic distances. Modern networks often combine these categories.

Exercise

Give one use case for each.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 31: OSI Reference Model

Lesson Content

The OSI model has seven conceptual layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Layering organizes responsibilities and helps troubleshooting and design.

Exercise

List the seven layers in order.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 32: TCP/IP Model

Lesson Content

The TCP/IP architecture is a practical protocol suite underlying Internet networking. A common teaching model uses Link, Internet, Transport, and Application layers. Variations exist depending on the reference used.

Exercise

Compare OSI and TCP/IP.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 33: Physical and Data Link Layers

Lesson Content

The physical layer covers media, connectors, signaling, and transmission of raw bits. The data link layer handles frames, local delivery, link-layer addressing, and often error detection.

Exercise

Differentiate the two layers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 34: Network and Transport Layers

Lesson Content

The network layer provides logical addressing and routing between networks. The transport layer provides end-to-end services. TCP provides reliable ordered delivery with flow and congestion control; UDP provides connectionless datagrams without TCP-style reliability.

Exercise

Compare TCP and UDP.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 35: Application Layer Protocols

Lesson Content

HTTP supports web communication; HTTPS protects HTTP with secure cryptographic transport; DNS provides naming and resource information; SMTP supports mail transfer; IMAP and POP support mailbox access in different ways.

Exercise

Name five application-layer protocols.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 36: IP Addressing Fundamentals

Lesson Content

IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses. IP addresses identify interfaces or endpoints within an IP network. Subnetting divides address space into smaller logical networks.

Exercise

Explain IPv4 versus IPv6.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 37: Ethernet and MAC Addresses

Lesson Content

Ethernet is a family of IEEE 802.3 LAN technologies. Ethernet frames contain addressing and control information. Switches learn source MAC addresses and forward frames toward appropriate destinations.

Exercise

Explain the relationship between frames, MAC addresses, and switches.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 38: Wireless Communication

Lesson Content

Wi-Fi is based on IEEE 802.11 technologies. Wireless systems can experience interference, attenuation, multipath effects, and congestion. Performance depends on distance, obstacles, channel conditions, device capability, and traffic load.

Exercise

List five factors affecting Wi-Fi performance.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 39: Network Devices

Lesson Content

Repeaters regenerate or retransmit signals. Traditional hubs repeat signals to ports. Switches forward local frames. Routers forward packets between networks. Access points connect wireless clients to a network.

Exercise

Differentiate these five devices.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 40: Communication Security

Lesson Content

Network security protects confidentiality, integrity, authenticity, and availability. Encryption protects data from unauthorized reading, authentication verifies identities, and secure protocols protect communications against relevant threats.

Exercise

Name three measures for protecting communication.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 41: Quality of Service

Lesson Content

QoS manages traffic characteristics such as delay, loss, jitter, and throughput. Interactive voice and video are often sensitive to delay and jitter. Classification and prioritization can provide appropriate treatment.

Exercise

Why might voice traffic need different treatment from a file download?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 42: Network Performance

Lesson Content

Throughput is the actual rate of useful data delivered. Latency is delay. Jitter is variation in packet delay. Packet loss occurs when packets fail to reach their destination. These metrics affect user experience and application behavior.

Exercise

Define the four performance terms.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 43: Client-Server and Peer-to-Peer

Lesson Content

In client-server systems, clients request services from servers. In peer-to-peer systems, peers can act as both clients and providers. Each architecture has different management, scalability, and availability characteristics.

Exercise

Compare client-server and peer-to-peer.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 44: Internet and Modern Networks

Lesson Content

The Internet interconnects networks using interoperable protocols, especially TCP/IP. Routing connects different networks. Cloud networking provides virtual network resources, while IoT connects physical devices and sensors.

Exercise

Explain how independent networks communicate over the Internet.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 45: Network Troubleshooting

Lesson Content

Troubleshooting should be systematic: identify symptoms, check physical connectivity, verify configuration, test connectivity, inspect logs, isolate the fault, apply a controlled fix, and document the result.

Exercise

Create a troubleshooting sequence for a PC with no Internet access.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 46: Network Design Case Study

Lesson Content

A fictional school needs administration, teacher, student, lab, server, Wi-Fi, and guest networks. Design logical separation, appropriate media, switching and routing, secure wireless, addressing, monitoring, and documentation.

Exercise

Draw the proposed topology and explain each component.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 47: Revision Guide

Lesson Content

Review communication components, signals, media, encoding, modulation, multiplexing, switching, error control, flow control, topologies, OSI/TCP-IP, addressing, Ethernet, wireless, devices, security, and performance.

Exercise

Explain how a message travels from a student's computer to a web server.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Final Assessment: 60 Questions and Answers

Attempt each question before reading the answer.

1. What is cybersecurity?

Answer: The practice of protecting systems, networks, devices, applications, and data from unauthorized or harmful activity.

2. What is an asset?

Answer: Something valuable that requires protection.

3. What is a threat?

Answer: A potential cause of harm.

4. What is a vulnerability?

Answer: A weakness that could contribute to a security incident.

5. What is risk?

Answer: The possibility and impact of an undesirable event.

6. What is a security control?

Answer: A safeguard used to prevent, detect, deter, or respond to security risks.

7. What does CIA stand for?

Answer: Confidentiality, Integrity, and Availability.

8. What is confidentiality?

Answer: Preventing unauthorized access to information.

9. What is integrity?

Answer: Protecting information from unauthorized or improper alteration.

10. What is availability?

Answer: Ensuring authorized access when needed.

11. What is malware?

Answer: Malicious software intended to perform unauthorized or harmful actions.

12. Name three malware categories.

Answer: Ransomware, spyware, and trojans.

13. What is phishing?

Answer: A deceptive attempt to trick people into revealing information or performing unsafe actions.

14. What is social engineering?

Answer: Manipulating people to bypass normal security behavior or reveal information.

15. Why use unique passwords?

Answer: To reduce the impact of one compromised password on other accounts.

16. What is MFA?

Answer: Authentication using two or more different factors.

17. What is least privilege?

Answer: Giving only the permissions necessary for a task.

18. What is role-based access control?

Answer: Assigning permissions according to defined roles.

19. What is network segmentation?

Answer: Separating networks or systems to limit unnecessary access and potential spread.

20. What is a firewall?

Answer: A control that filters network traffic according to rules.

21. Why secure Wi-Fi?

Answer: To prevent unauthorized access and protect network communications.

22. What is encryption?

Answer: Transforming data into protected form using a cryptographic method and key.

23. What is symmetric encryption?

Answer: Encryption using the same secret key for encryption and decryption.

24. What is asymmetric cryptography?

Answer: Cryptography using related public and private keys.

25. What is hashing?

Answer: A one-way transformation that produces a fixed-length digest of data.

26. Why patch systems?

Answer: To address known vulnerabilities and improve security.

27. What is web security?

Answer: Protecting web applications, services, sessions, data, and users from relevant threats.

28. Why is input validation important?

Answer: It helps ensure application input is handled safely and as expected.

29. What is a security policy?

Answer: A formal statement of security requirements and expected behavior.

30. What is security awareness?

Answer: Education that helps people recognize and respond appropriately to security risks.

31. What is vulnerability management?

Answer: Identifying, prioritizing, remediating, and tracking security weaknesses.

32. What is security monitoring?

Answer: Collecting and analyzing security-relevant events to identify problems or suspicious activity.

33. What is incident response?

Answer: A structured process for detecting, containing, investigating, recovering from, and learning from security incidents.

34. Why preserve evidence?

Answer: To maintain its reliability and support accurate investigation.

35. What is chain of custody?

Answer: Documentation of who handled evidence, when, why, and how.

36. Why are backups important?

Answer: They allow restoration after loss, corruption, hardware failure, or certain security incidents.

37. What is RTO?

Answer: Recovery Time Objective: a target for restoring a service.

38. What is RPO?

Answer: Recovery Point Objective: the amount of recent data loss that may be acceptable.

39. What is secure software development?

Answer: Integrating security into requirements, design, coding, testing, deployment, and maintenance.

40. What is threat modeling?

Answer: Systematically identifying assets, threats, trust boundaries, and controls during design.

41. Why protect secrets?

Answer: Exposed credentials can provide unauthorized access to systems and data.

42. What is cloud shared responsibility?

Answer: Security responsibilities are divided between the provider and customer according to the service and configuration.

43. Why protect databases?

Answer: They often contain valuable or sensitive records.

44. What is data minimization?

Answer: Collecting and retaining only information necessary for legitimate purposes.

45. What is defense in depth?

Answer: Using multiple layers of complementary security controls.

46. What is Zero Trust?

Answer: A model emphasizing verification and least privilege rather than implicit trust.

47. What is endpoint security?

Answer: Protecting user devices such as computers and phones.

48. What is IAM?

Answer: Identity and Access Management.

49. Why review permissions?

Answer: To detect excessive, outdated, or inappropriate access.

50. What is a SOC?

Answer: A Security Operations Center that may coordinate security monitoring and response.

51. What is a playbook?

Answer: A documented set of response steps for a recurring security event.

52. Why is security training important?

Answer: It helps users recognize risks and follow safer procedures.

53. What is a dark pattern?

Answer: A deceptive interface that manipulates users into unintended actions.

54. Why is privacy part of security?

Answer: Security controls help protect personal information from unauthorized exposure or misuse.

55. What is a risk register?

Answer: A structured record of identified risks, impacts, controls, owners, and treatment decisions.

56. What is an asset inventory?

Answer: A record of systems, devices, applications, and other assets that need protection.

57. Why is logging important?

Answer: Logs provide information for monitoring, investigation, troubleshooting, and accountability.

58. Why test backups?

Answer: To verify that data can actually be restored when needed.

59. What is the most important rule for security testing?

Answer: Only test systems when you have explicit authorization and remain within the defined scope.

Practical Exercises

Exercise 1 — Communication Model

Choose a video call, email, web request, or file transfer. Identify sender, receiver, message, medium, and protocol.

Exercise 2 — Media Comparison

Create a table comparing twisted pair, coaxial cable, fiber, radio, microwave, and infrared using distance, bandwidth, interference resistance, cost, and applications.

Exercise 3 — Network Topology

Draw star, bus, ring, and mesh topologies. State one advantage and one limitation for each.

Exercise 4 — OSI Mapping

Take a web request and identify example responsibilities and protocols at each relevant layer.

Exercise 5 — Troubleshooting

Create a diagnostic procedure for a PC that is connected to Wi-Fi but cannot reach a website. Include physical, addressing, DNS, routing, and application checks.

Data Communication Glossary

Accessibility: Designing products for people with diverse abilities.

Affordance: What an object or interface element can support.

Breadcrumb: Navigation showing a user's position in a hierarchy.

Component: Reusable interface element with defined behavior.

Design system: Reusable design assets, rules, components, and documentation.

Empathy map: A tool for organizing user statements, thoughts, actions, and feelings.

Fidelity: How closely a prototype represents the intended final experience.

Flow: A sequence of steps through a product to accomplish a task.

IA: Information architecture.

Interaction design: Design of actions, responses, states, and behavior.

Persona: Research-informed representation of a user group.

Prototype: Model used to explore or test a design.

Responsive design: Design that adapts to available space and context.

Usability: How effectively, efficiently, and satisfactorily users achieve goals.

User journey: Description of an experience across stages and touchpoints.

User research: Systematic study of users, behavior, needs, and context.

Wireframe: Simplified representation of interface structure.

UX writing: Content designed to guide users through an interface.

End of Data Communication Fundamentals

Data communication is the foundation of computer networking. Strong understanding of signals, media, protocols, addressing, switching, error control, and performance provides the foundation for advanced networking and cybersecurity study.